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From literature to research-based learning: An AI-powered information extraction system to enhance undergraduate thesis completion

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ABSTRACT

Undergraduate thesis completion in STEM education is often constrained by limited timeframes, insufficient training, and the difficulty of navigating unstructured scientific literature. This study describes the design and evaluation of an AI-powered information extraction system that converts research publications into structured, traceable, and comparable datasets to support literature-driven comparison. The system is positioned as an epistemic scaffold that enables the inspection of evidence claim relationships through source traceability and structured comparisons while reducing low-level data-handling demands. A mixed-methods pilot study was conducted with 20 undergraduates from four schools and four academic supervisors within the same university, combining system log analysis, post-task surveys, and semi-structured interviews. Across 80 uploaded documents, students successfully extracted and standardized over 90% of the targeted experimental parameters and performance entries. Students' self-reported literature review time decreased by approximately 65%, and their ability to identify at least three influential variables in experimental designs increased by 50% in a brief pre/post variable-identification prompt. Survey and interview findings indicated that the structured outputs enabled students to transition from isolated reading toward cross-study comparison and evidence-based justification. Meanwhile, supervisors reported a reduced need for repeated instruction and improved students' reasoning capacity. The pedagogical implications for data-informed supervision are discussed, along with limitations related to sample size, domain specificity, and the need to mitigate overreliance on automation through explicit verification routines. The modular architecture of the system suggests its transfer potential to other STEM domains in which parameter-based comparison is central. Given the exploratory, single-institution pilot design, quantitative results are interpreted as descriptive and mechanism-oriented rather than statistically generalizable.

1. Introduction

The undergraduate thesis, often referred to as a capstone course or final-year project, is widely adopted in STEM education as a culminating academic experience that integrates disciplinary knowledge with inquiry, data analysis, and scientific communication (Carter, 2011; Sastry et al., 2010; Tenhunen et al., 2023). Empirical studies suggest that this final-year research experience can strengthen students' critical thinking, academic writing, and ability to justify claims with evidence (Green et al., 2024; Henttonen et al., 2023; Shaw et al., 2013). However, the literature also reports persistent barriers to undergraduate thesis completion, including limited research preparedness, constrained timelines, and academic supervisor workload, which can restrict students' epistemic engagement with primary literature and reduce analysis depth (Wissinger, 2018).

The allocated duration for the undergraduate thesis varies across

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higher education systems. In Japan, undergraduate students typically have one full academic year to complete their thesis, often starting at the beginning of their final year. By contrast, universities in other Asian countries, such as South Korea, Singapore, and China, generally assign only one semester for undergraduate thesis completion, roughly 5-6 months, typically from January to May. Similarly, undergraduate theses in the United States are often completed during the final spring semester of a four-year program. In Australia, undergraduate students undertaking honors or thesis-track degrees usually have between 6 months to one year for thesis completion, although the latter option is not universally adopted. In Europe, undergraduate thesis timelines range from 2 to 4 months. Such tight deadlines are challenging for students, especially those engaging in research for the first time.

Despite structural variations between universities, the majority of undergraduate thesis programs provide students with only a few months to complete their independent research project, often with minimal training in topic selection, critical literature review, performing experiments or simulations, data analysis, and thesis writing (Carter, 2011; Delaney, 2011; Tenhunen et al., 2023; Zdravkova & Ilijoski, 2025). Superficial outcomes are often achieved when students are expected to perform a genuine scientific investigation within a limited timeframe, especially for those without prior exposure to research (Mann et al., 2020; Roberts & Seaman, 2018; Wissinger, 2018). Such students struggle to thoroughly review the state of the art in their field, propose genuinely novel research questions, or gather and analyze sufficient empirical data. Additionally, many senior undergraduate students are preoccupied with preparing for graduate entrance examinations or job hunting, further dividing their attention and cognitive resources. Moreover, academic supervisors often face heavy workloads (Weaver et al., 2016), typically guiding several students at once, while being expected to provide timely, individualized feedback and avoid one-size-fits-all guidance. This commitment is difficult to achieve under real-world teaching constraints.

The literature review stage is one of the most substantial bottlenecks in completing undergraduate thesis work within a few months (Wissinger, 2018). For example, an undergraduate materials science student attempting to explore the properties of a specific class of advanced nanomaterials may find quickly identifying and comparing existing research findings exceedingly difficult due to the volume, complexity, and heterogeneity of published studies (Gupta et al., 2022; Huang & Cole, 2022). From an educational perspective, this stage is critical for the development of epistemic cognition, i.e., students' understanding of how knowledge is constructed and validated, yet many undergraduate students lack the necessary scaffolds to effectively make this transition (Gao et al., 2025). The inability to efficiently extract and organize relevant data across a rapidly expanding research field, not only hampers students' ability to define a clear research direction, but also places a substantial burden on their academic supervisors.

Existing research on digital technologies (Ramli & Mahmud, 2025; Smerdon, 2024) has highlighted the potential of AI to support inquiry-based learning and research-related practices in higher education, including literature engagement and data reasoning. However, much of this work has focused on general-purpose AI assistance (e.g., summarization or conversational support), or tool evaluations in single courses. Consequently, limited evidence supports the use of structure-aware AI systems to perform cross-document information extraction and normalization in undergraduate thesis work. Moreover, little is known about how literature-to-data process traces can be aggregated into pedagogically interpretable indicators to support formative supervision.

To address this knowledge gap, this study describes the design, implementation, and pedagogical evaluation of an AI-powered information extraction system for undergraduate thesis work in STEM education. The innovation of the proposed system lies not only in the tool itself but in an educational approach that reframes the literature review as a structured, data-driven learning activity and embeds AI-supported extraction into undergraduate thesis supervision. This approach enables process-oriented guidance under compressed timelines and supports students' transition from passive reading to evidence-based reasoning. Co-developed by undergraduate students and supervisors, the proposed system transforms unstructured scientific literature into structured and comparable datasets. Theoretically, the system builds upon research in inquiry-based learning, data literacy, and constructivist pedagogy, positioning AI not as a shortcut but as a scaffold for epistemic cognition development. Specifically, epistemic cognition is supported through traceable source links and cross-study comparison, cognitive load is reduced by automating search/transcription/standardization steps, and digital scaffolding is operationalized through guided extraction templates and supervisor-facing process indicators. From a practical perspective, the system enables students to complete meaningful thesis work within tight deadlines while reducing supervisory burden. The contributions of this study to educational technology and AI-in-education literature are threefold: (1) it provides a conceptually grounded account of structure-aware extraction as an epistemic scaffold for literature-driven inquiry; (2) it presents a process-oriented supervision model enabled by traceable intermediate artifacts, including extracted variables, comparison tables, source links, and workflow traces; and (3) it describes a mixed-methods evaluation that triangulates learning outcomes, user perceptions, and descriptive analytics as process evidence of research-based learning engagement.

2. Theoretical framework

This study integrates three complementary educational lenses to frame the system architecture and results interpretation: (1) Epistemic cognition informs how thesis-based learning advances when students evaluate evidence claim relationships. Accordingly, the system emphasizes traceable sources and structured comparison. (2) Cognitive load theory helps to explain why automating repetitive search, transcription, and unit normalization can reduce extraneous information processing and free cognitive resources for analysis and justification. (3) Digital scaffolding guides the design principle that AI should support learners' sense-making and verification rather than replace disciplinary reasoning.

2.1. Scientific literacy and the undergraduate thesis

Scientific literacy (Eslinger & Kent, 2018), defined as the ability to interpret, evaluate, and apply scientific information, is a fundamental competency for research engagement in higher education. In the context of undergraduate thesis completion, scientific literacy enables students to understand prior work, identify research gaps, and construct evidence-based arguments (Reynolds & Thompson, 2011). However, for many students, particularly those in STEM domains, the undergraduate thesis is their first formal encounter with independent research. Students lacking experience in critically reading literature and synthesizing and comparing data across studies often struggle to acquire the scientific literacy needed to perform meaningful research, especially within limited timelines. This stage can also be understood as a critical moment in the development of epistemic cognition, which is widely recognized as being central to research-based learning in STEM education.

Ideally, the undergraduate thesis is a vehicle through which students transition from passive recipients of knowledge to active participants in disciplinary inquiry (Mann et al., 2020; Medaille et al., 2022). However, the increasing complexity and volume of scientific publications have created a high barrier to research-based learning. Nevertheless, studies have reported that epistemic cognition can substantially evolve when undergraduate engineering students are placed in authentic research or mentoring roles (Gao et al., 2025). This highlights the importance of structured support in shifting students from passive knowledge receivers to active knowledge constructors. A material information extraction system that supports students in quickly navigating field-specific literature and understanding data trends (Gupta et al., 2022; Hira et al., 2024; Huang & Cole, 2022) is urgently needed to not only improve thesis outcomes but develop long-term research capacity.

2.2. Data literacy and the need for structured information access

Closely related to scientific literacy is data literacy, which refers to the ability to locate, interpret, and reason with data in diverse formats. Students must be able to extract experimental data, performance metrics, and contextual information from a wide range of publications during undergraduate thesis work. However, such data are often embedded within heterogeneous textual formats, tables, and figures, without standardized representation (Gupta et al., 2022; Huang & Cole, 2022), posing a considerable burden on students with limited training. From the perspective of *cognitive load theory* (Gkintoni et al., 2025; Paas & Van Merriënboer, 1994; Sweller, 1988; van Merriënboer & Sweller, 2005), the sheer volume and inconsistency of research data hinder students' ability to focus on essential disciplinary reasoning. AI-powered systems can reduce this extraneous cognitive load by transforming unstructured text into structured datasets, freeing their cognitive resources for deeper learning and analytical tasks.

A key challenge for students performing literature reviews in undergraduate research is not the lack of data, but the lack of structured and comparable data (Round & Campbell, 2013; Wissinger, 2018). This challenge is particularly acute in fields such as chemical engineering, materials science and engineering, mechanical engineering, bioengineering, chemistry and physics, where performance depends on experimental conditions that vary across studies (Gupta et al., 2022; Hira et al., 2024; Huang & Cole, 2022). Students' ability to define research questions or conduct meaningful analysis is severely limited when they cannot efficiently extract and compare such information. Thus, developing data literacy in tandem with scientific literacy is not only a technical necessity but also a pedagogical imperative in STEM education (Kjelvik & Schultheis, 2019).

2.3. The role of AI in research-based learning

AI has emerged as a transformative tool in educational contexts, and recent advances have enabled the development of systems capable of extracting, classifying, and structuring information from large volumes of text (Chan & Hu, 2023; Liu et al., 2024; Roski et al., 2024; Timotheou et al., 2023). In research-based learning, AI can assist students in identifying key concepts, summarizing the literature, and even generating structured representations of performance data, thereby lowering the barrier to research engagement (Song et al., 2024). Importantly, AI in this context is best understood as a digital scaffold, i.e., a tool that amplifies students' reasoning capacity without replacing it (Kasneci et al., 2023; Kim et al., 2022), aligning with the tradition of inquiry-based learning where learners construct meaning through exploration and evidence evaluation (Pedaste et al., 2015).

Employing semantic mining and entity extraction techniques to convert literature data into structured and comparable datasets can support students in the literature review phase by substantially reducing the time needed to understand research trends, identify relevant studies, and form comparative insights (Gupta et al., 2022; Hira et al., 2024; Huang & Cole, 2022). Moreover, AI-powered systems can function as epistemic scaffolds that guide students toward recognizing how knowledge is generated and validated within their domain (Xi et al., 2026).

2.4. Pedagogical paradigm: From tool to scaffolding system

From an educational technology perspective, the proposed AI-powered information extraction system represents a paradigm shift from a static digital tool to a dynamic instructional scaffold that supports learners during complex, open-ended research tasks. Consistent with research on scaffolding in inquiry learning (Belland, Walker, & Kim, 2017; Reiser, 2004), the system in this study provides structured supports, such as guided extraction templates, traceable source links, and comparison views to help novice researchers carry out literature-based reasoning that would otherwise be difficult to accomplish. This approach reflects the principles of constructivist pedagogy, where learning is not merely the accumulation of knowledge, but the construction of knowledge through inquiry, exploration, and comparison (Dennick, 2016; Taber, 2024). However, evidence on technology-mediated scaffolding is not uniformly positive, and learning benefits are contingent on design and implementation. Meta-analytic evidence on computer-based scaffolding in STEM education indicates heterogeneous effects and highlights that outcomes vary with scaffolding features, learner characteristics, and instructional integration (Belland, Walker, & Kim, 2017). Recent evidence also cautions that students may develop over-reliance on AI systems, accepting AI outputs without sufficient verification, which can undermine critical cognitive abilities such as analytical reasoning and decision-making (Zhai et al., 2024). This motivates our design stance: the extraction system should be embedded in explicit instruction and source-verification routines, so that AI support strengthens students' epistemic agency rather than encouraging automation dependence.

By integrating epistemic cognition, cognitive load theory, and digital scaffolding, the system in this study is positioned not only as a technical tool but as an educational scaffold that enables efficient undergraduate thesis completion while cultivating transferable STEM competencies such as data reasoning, hypothesis generation, and critical evaluation. Based on theoretical framing and the identified research gap, this study addressed the following research questions (RQs):

(RQ1) To what extent does the system improve students' efficiency and accuracy in literature-to-data tasks (e.g., extraction coverage/accuracy and time-on-task) and their ability to identify key variables?
(RQ2) How do students and supervisors perceive the usability of the system and its role as a scaffold for research-based learning and supervision?
(RQ3) What do the learning analytics traces reveal about students' engagement and workflow progression, and how can this inform formative supervision?

3. Method

To directly address the challenges of undergraduate thesis completion, particularly difficulties of navigating complex literature, extracting performance-relevant data, and constructing evidence-based comparisons under tight deadlines, the proposed AI-powered information extraction system was designed to transform raw, unstructured academic content into structured, queryable and comparable knowledge.

3.1. Research design

Guided by the research questions stated in Section 2.4, this study employed a mixed-methods case study design, combining system log analysis, post-task surveys, and semi-structured interviews, to capture system use traces and participants' subjective experiences. The case study involved 20 undergraduate students from four schools within the same university (Table 1), including four senior students who were conducting their undergraduate theses in materials science and 16 junior students from other disciplines who acted as novice testers without prior knowledge of the thesis topic. Four supervisors from the same institution were also involved in guiding senior students and supporting junior students. The study design enabled observations of the impact of the system across both experienced and inexperienced learners, as well as the supervisor perspective.

Participant selection followed a purposive and convenience-based

Table 1
Participant backgrounds and thesis titles/focus areas.

Student No.	Level	School	Experience with thesis topic	Thesis title/focus area
1	**Senior**	**SMSE** [a]	**Yes**	**Nanofriction of fluorine-free ionic liquids under extreme loads**
2	Senior	SMSE	Yes	**Viscosity variation of nonhalogenated ionic liquids at extreme temperatures**
3	Senior	SMSE	Yes	**Nanofriction properties of phosphonium-based ionic liquids confined on graphite surfaces**
4	Senior	SMSE	Yes	**Structure-property relationships of confined imidazolium-based ionic liquids at solid interfaces**
5 **6** **7** **8**	Junior	SMSE	**No**	**Focusing on the same topic as Student 1**
9 **10** **11** **12**	Junior	**SEM** [b]	No	Focusing on the same topic as Student 2
13 **14** **15** **16**	Junior	**SCCE** [c]	No	Focusing on the same topic as Student 3
17 **18** **19** **20**	Junior	**SIP** [d]	No	Focusing on the same topic as Student 4

[a] SMSE, School of Materials Science and Engineering.
[b] SEM, School of Economics & Management.
[c] SCCE, School of Chemistry and Chemical Engineering.
[d] SIP, School of Intellectual Property.

sampling strategy that was appropriate for an exploratory mixed-methods case study. The four senior students comprised the complete set of final-year thesis students supervised by the first author in the 2024 cohort with thesis topics that aligned with ionic liquid lubrication. This selection of participants enabled the authentic integration of the extraction system into an ongoing thesis workflow without disrupting assessment arrangements. The 16 junior students were recruited via invitation to serve as novice testers without prior domain knowledge, allowing the examination of onboarding, usability, and scaffolding effects for non-expert learners.

Data sources and analyses were aligned with the research questions. RQ1 was answered by assessing extraction coverage/accuracy and time-on-task indicators from the system logs and learning analytics dashboard, together with a pre/post variable-identification prompt in interviews. RQ2 was answered through post-task surveys and interview accounts from students and supervisors. RQ3 was answered by assessing learning analytics dashboard indicators, such as the number of documents processed, extraction completion counts, extraction iterations, search actions, and daily activity trends, to characterize engagement and progression through the ingestion–extraction–comparison–thesis application workflow.

3.2. System design and workflow

The system integrated state-of-the-art information extraction techniques with an educationally responsive interface, ensuring technical scalability and instructional usability. Fig. 1 illustrates the end-to-end process from document ingestion, preprocessing, and semantic field extraction to structured output and iterative refinement. Yellow dashed boxes indicate customizable modules that can be modified to suit different research domains, enabling the system to be repurposed for interdisciplinary research-based learning with minimal redevelopment effort. The system architecture integrated fine-tuned models, manual validation loops, and domain-adaptable visualization, supporting technical adaptability and educational scalability. At the time of this study, we have not yet completed a full deployment in a different domain beyond materials science. System modularity was operationalized in the pilot study through template-level reconfiguration across four thesis topics within the same cohort (Table 1). This operation required

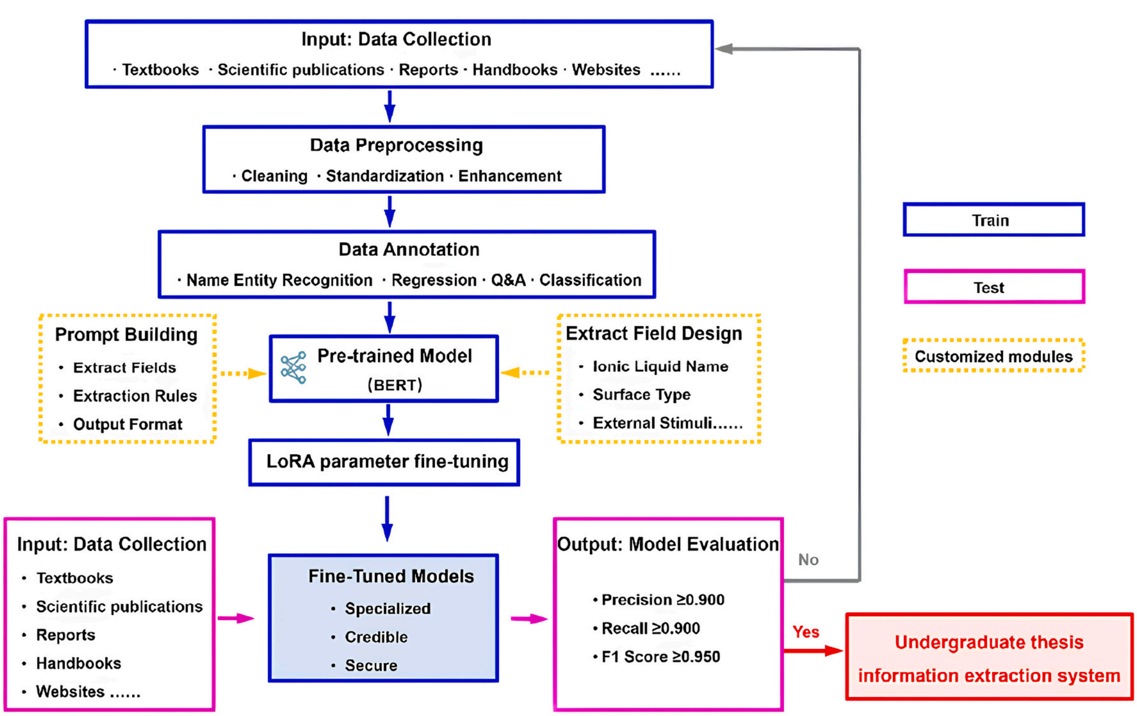

Fig. 1. Workflow of the AI-powered information extraction system.

adjusting target variables (e.g., friction coefficient or viscosity), condition fields (e.g., load or temperature ranges), and normalization rules, while keeping the core pipeline (ingestion–extraction–standardization–database–comparison/visualization) unchanged. This within-domain porting provided process evidence that the customizable modules in Fig. 1 could be reconfigured with minimal redevelopment effort and clarified the current boundary of empirical evidence for system transferability.

The end-to-end process comprised the following steps: 1) Corpus ingestion: Students upload relevant documents including textbooks, handbooks, peer-reviewed journal articles, technical reports, and relevant websites that are commonly used during early-stage topic exploration and literature review. Notably, the system was not intended to replace bibliographic search engines (e.g., Web of Science), but to operate downstream of literature discovery by ingesting full-text sources and transforming them into structured, parameter-level data for comparison, traceability, and reuse. 2) Preprocessing: The uploaded documents undergo preprocessing for section segmentation, figure–text integration, and metadata alignment, helping students to focus on the content most relevant to their research design (e.g., Materials and Methods or Results and Discussion) and avoid being overwhelmed by entire documents. Bulk upload functionality allowed students to process dozens of studies in parallel, enabling them to effectively explore research landscapes efficiently. 3) Semantic field extraction: Students extracted key entities (e.g., friction coefficient, load, temperature, surface type, and ionic liquid category) through custom prompts and predefined field templates (Fig. 2). 4) Structured output and visualization: The extracted results are standardized, validated, and stored in a database, enabling students to compare cross-study findings through tables, filters, and interactive charts (Fig. S1–S2). 5) Iterative refinement: Extraction models are continuously retrained with new documents and entries are manually verified/edited by users, ensuring traceability and evolving accuracy.

As shown in Fig. 2, the "Advanced Search" interface (upper panel) enables targeted retrieval of literature-derived results. The Standard Search mode (bottom panel) provides a broader query option, listing all extracted entries that match a given keyword, together with experimental conditions and reference links. The "Confirm" hyperlink in the Reference Details panel opens the original source article, enabling students to verify the extracted data in context. The extracted data are then standardized, validated, and stored in a structured database, enabling the direct comparison of experimental conditions and performance metrics across studies. This structured comparison mirrors the way experienced researchers conduct state-of-the-art reviews, making expert practices accessible to novice researchers. The "Reference Details" and "Edit Data" windows (marked by dashed boxes in Fig. 2) support traceability by linking each entry to its original publication, with precise source location (page and table number), and allow manual corrections or updates when necessary. Each extracted record is stored with the bibliographic metadata of the original publication (e.g., title and DOI), and an explicit traceability pointer to the evidence location (e.g., page and table/figure). The primary literature reporting extracted values must be cited rather than the platform output. In exported or compiled comparison tables, each row retains its corresponding reference field, enabling one-to-one mapping between reported values and citations.

3.3. Pedagogical alignment

The AI-powered information extraction system was explicitly designed to align with the four typical stages of undergraduate research:

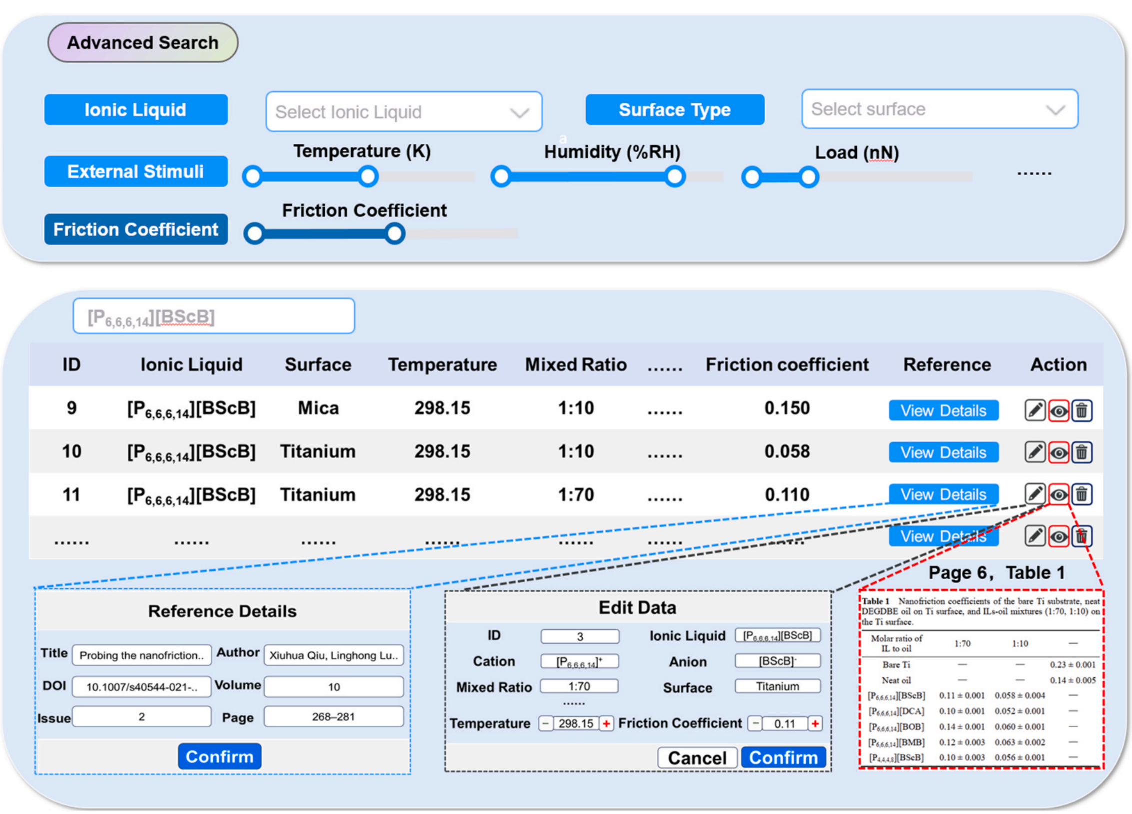

Fig. 2. Representative user interface for retrieving and managing structured literature data in the AI-powered information extraction system (a case study of undergraduate thesis on ionic liquid lubrication). The upper panel shows the "*Advanced Search*" function, enabling multi-parameter filtering (e.g., ionic liquid type, surface type, temperature, humidity, load, and friction coefficient) to locate specific experimental conditions in the database. The bottom panel displays the *standard search* results table with extracted parameters, measured friction coefficients, reference links, etc., with the panels marked by dashed boxes illustrating the "Reference Details" window (left), "Edit Data" window (middle), and raw data (right).

topic exploration and literature review, experimental design and planning, data analysis and comparison, and writing and discussion of results. Typically, engineering students would be able to identify preliminary structure–property relationships by comparing the performance data of different materials under varying experimental conditions extracted from the literature. This comparative analysis helps them to develop a more holistic understanding of the research landscape relevant to their thesis topic (e.g., key variables, trends, and methodological approaches) and better prepares them for the transition to the next research stage with greater clarity, direction, and confidence, moving beyond superficial reading toward data-informed interpretation. This pedagogical alignment ensured that the AI-powered system acted as a scaffold for epistemic cognition, making expert practices accessible to novice researchers and reducing the need for repeated instruction. This alignment constitutes an educational innovation by operationalizing thesis supervision as a structured learning design, with AI-mediated research artifacts enabling formative, process-based guidance, rather than relying solely on end-product evaluation.

To maintain students' epistemic agency and ensure transferable data extraction competency, system use was paired with an explicit data extraction literacy and source verification routine. In the onboarding session, students practiced manual extraction using a structured worksheet that included the target variable, unit, boundary conditions, and page/table location, and then students compared their entries with the extraction system output. Using the "Reference Details" and raw data views (Fig. 2), students could inspect how each extracted entry was grounded in the original source and correct entries through the editable interface when mismatches occurred. During the pilot study, students were asked to verify the key values they planned to cite or use in their experimental design, treating the extraction system output as a starting point for critique rather than unquestioned answers.

3.4. Data collection

Participants completed a brief onboarding activity via a recorded screencast video hosted on the platform prior to using the AI-powered information extraction system. The video demonstrated (i) the end-to-end workflow of the system for literature ingestion, field configuration, extraction, and structured output; (ii) the corresponding manual procedure for extracting the same variables directly from the original papers including locating values in text/tables/figures and recording units; (iii) how to cross-check and validate system-extracted entries against manual extraction using a verification checklist; and (iv) how to correct, annotate, and update erroneous entries in the database when discrepancies were identified. This onboarding process was designed to ensure that students understood the underlying extraction logic and were able to independently verify and curate the extracted results, thereby mitigating overreliance on automation.

Three complementary data sources were employed: 1) system log data including students' workflow traces (document ingestion, search/filter actions, time-on-task [e.g., average extraction time], success rate, and daily activity trends), which were summarized in the learning analytics dashboard (see screenshot in Fig. S3) and utilized to quantify efficiency gains and extraction coverage and serve as process evidence of workflow progression; 2) post-task surveys which were administered to all students and supervisors, focusing on ease of use, learning support, research understanding, and supervision experience; and 3) semi-structured interviews, which were conducted with all students and supervisors, probing the influence of the system on students' literature comprehension, experimental reasoning, and confidence in research tasks. Supervisors were asked about perceived changes in students' autonomy, engagement, and workload. While surveys and interviews were administered post-task, the platform also captured in-task behavioral traces via system logs and dashboard indicators, and juniors completed a brief pre/post variable-identification prompt to provide a within-participant descriptive comparison.

The selection of these instruments was guided by the research questions and the need to triangulate objective process measures with participants' perceptions and explanations. First, system logs and Learning Analytics Dashboard indicators are automatically captured by the platform, reducing recall bias and improving measurement consistency. To support construct validity, each indicator was defined operationally and interpreted as a process trace of learners' progression through the ingestion-extraction-comparison-thesis application workflow (RQ1, RQ3). Second, the post-task surveys were designed to reflect the key evaluation dimensions (ease of use, learning support, research understanding, supervision experience) on educational technology and AI-supported learning. Items were reviewed and refined for clarity and relevance by the research team and participating supervisors, and the results are interpreted as descriptive evidence to complement the logs and interviews (RQ2). Third, semi-structured interviews were used to elicit concrete examples of how participants used traceability, comparison, and verification when making research decisions, thereby strengthening explanatory depth and supporting methodological triangulation (RQ2).

3.5. Data analysis

3.5.1. Quantitative analysis

System log data and Learning Analytics Dashboard indicators were exported from the platform as tabular files (e.g., CSV) and summarized using descriptive statistics (counts, proportions/percentages, and time summaries). All calculations and figures were produced using standard spreadsheet/statistical scripting tools (e.g., Excel and/or equivalent data-processing software) to ensure transparent and reproducible aggregation of log indicators. Given the exploratory case-study design and the small sample ($N = 20$), we did not conduct inferential hypothesis testing and interpret quantitative results as descriptive/process evidence rather than statistically generalizable effects. Accordingly, we report distributions and percentage-point changes and avoid inferential claims, recognizing that individual differences may influence aggregated descriptive patterns in a small-N pilot.

Extraction coverage was computed as: coverage = (number of uploaded documents for which at least one target field was successfully extracted)/(total uploaded documents). Extraction accuracy was estimated via manual verification against the original sources using traceability pointers (page/table/figure) in the system. Specifically, 20% of extracted entries was cross-checked by comparing the extracted value and unit-normalized form with the evidence location in the source document; accuracy was computed as: accuracy = (number of verified-correct entries)/(total verified entries).

Time-on-task indicators (e.g., average extraction time and daily activity trends) were derived directly from log timestamps and were treated as engagement/workflow traces rather than achievement scores. Post-task survey responses were summarized at the item level using proportions/percentages to describe perceived usability and learning support. The pre/post "variable-identification" outcome reported in Results was based on a structured interview prompt in which students listed influential variables in experimental design; responses were tallied by the number of distinct variables identified and then grouped into 0-1, 2, 3, and >4 categories for visualization. Because the study is not powered for inferential modeling, changes are reported descriptively (e.g., percentage-point differences) and triangulated with qualitative evidence.

3.5.2. Qualitative analysis

Interview data were analyzed using a structured content analysis approach aligned with the interview protocol. Specifically, interview responses were first organized into the five evaluation dimensions defined a priori by the instrument (ease of use, learning support, research understanding, supervision experience, and improvement suggestions). Within each dimension, we iteratively reviewed the documented interview responses to identify recurring perceptions, concrete examples of system use (e.g., traceability checking, cross-study comparison, supervisor feedback routines), and points of divergence between participants. The resulting within-dimension themes were used to structure the qualitative reporting in the section of Results and to triangulate interpretations with the post-task survey results and the system log/learning analytics indicators. To maintain a critical stance, we also retained cautionary reflections and divergent accounts (e.g., perceived risks of overreliance and reminders about verification) rather than only summarizing positive experiences.

3.6. Ethical considerations

All participants provided informed consent prior to data collection. Data were anonymized, and participation had no impact on academic evaluation.

4. Results

A pilot study was conducted to evaluate the real-world applicability and pedagogical value of the AI-powered information extraction system. The four senior students had already selected thesis topics related to the lubrication performance of ionic liquids under extreme conditions and possessed basic background knowledge (Table 1). In parallel, 16 junior students from four different schools (Table 1) were invited to test the system without prior knowledge of ionic liquid lubrication. This dual-cohort design facilitated an evaluation of the ability of novice and semi-experienced students to effectively use the system.

All students began by uploading curated sets of scientific documents, including journal articles, reviews, patents, and reports, and extracted key data using the field configuration interface. The system returned structured tables with normalized units, source links, and visualization tools, enabling students to identify frequently studied combinations, discern dominant experimental conditions, and locate performance data across publications.

The results are organized around the research questions. Section 4.1 addresses RQ1 and RQ3 using system logs and learning analytics dashboard indicators. Sections 4.2 and 4.3 address RQ2 by summarizing the survey and interview findings. Section 4.4 synthesizes the observed impacts on research-based learning.

4.1. System log analysis

Students followed a typical four-step loops including document ingestion, extraction field configuration, output comparison and interpretation, and thesis application. Compared with previous student cohorts, supervisors reported a reduced need for repeated instruction and greater clarity in students' experimental logic. The quantitative evaluation further supported the utility of the system. Among the 80 uploaded documents, over 90% of relevant data entries were successfully extracted and structured (Fig. 3). Students' literature review time decreased by approximately 65% (self-reported in follow-up informal interviews; Table S1). Among the 16 junior students, the proportion who could correctly identify at least three influencing variables for experimental design increased from 37.5% (6/16) to 87.5% (14/16) after system use (Fig. 4), based on follow-up informal interviews (Table S1). We focused this pre/post comparison on juniors to avoid confounding effects from seniors' prior domain experience. Summaries of extraction coverage, topic comprehension improvement, and system usage over time are presented in Figs. 3 and 4. Results demonstrated that an intelligent information extraction system can serve not only as a research accelerator but also as a pedagogical scaffold that supports undergraduate students in navigating, understanding, and applying field-specific knowledge in a structured and meaningful way.

Subsequently, the learning analytics dashboard indicators were assessed to more thoroughly evaluate the educational and pedagogical

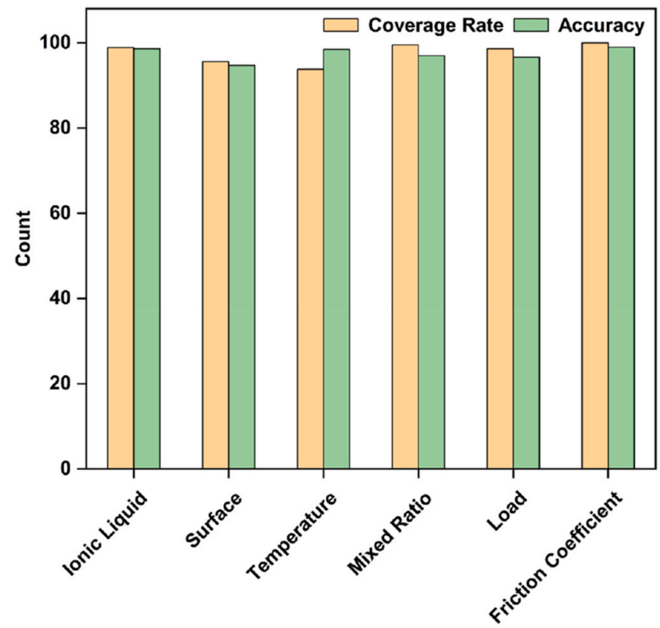

Fig. 3. Data extraction coverage and accuracy achieved by the information extraction system across 80 uploaded documents. The coverage rate represents the proportion of documents for which target data were successfully identified and extracted. Accuracy reflects the proportion of correctly normalized entries, validated through manual cross-checking.

implications of the system. The learning analytics dashboard summarized the number of documents processed, extraction completion counts, extraction iterations, search actions, average extraction time, daily activity trends, and extraction-source distributions for a specified time window. These indicators provided evidence of students' engagement and workflow progression, enabling supervisors to identify bottlenecks, such as repeated iterations or unusually long extraction times, and provide timely feedback. Notably, these indicators were interpreted as pedagogically meaningful traces of research-based learning rather than as purely technical performance metrics.

4.2. Survey results

Post-task surveys with all students and supervisors revealed consistently positive evaluations of the AI-powered information extraction system (Fig. 5). As shown in Fig. 5a, over 90% of students agreed or strongly agreed that the system improved the efficiency and clarity of their literature review process, reducing the time needed to collect and compare experimental data from multiple sources. Approximately 85% of students highlighted that the structured tables and visual filters helped them to identify key experimental variables and performance trends within large datasets, while more than 80% of students considered the interface intuitive and easy to use. Students also reported that the visualization functions enhanced their understanding of relationships among parameters, allowing them to transition from isolated reading toward comparative reasoning.

Similarly, supervisors gave favorable evaluations of the AI-powered information extraction system (Fig. 5b). All supervisors reported that the system reduced the need for repeated instruction and improved the transparency of students' research logic. Moreover, 75% of supervisors agreed that the platform facilitated differentiated guidance, as it allowed students to present filtered data and preliminary comparisons prior to consultation. All supervisors expressed interest in institutionalizing the system as part of the undergraduate thesis preparation curriculum, citing its potential to improve supervision efficiency and students' autonomy. Students and supervisors indicated that the system could be further improved by adding context-sensitive tooltips for

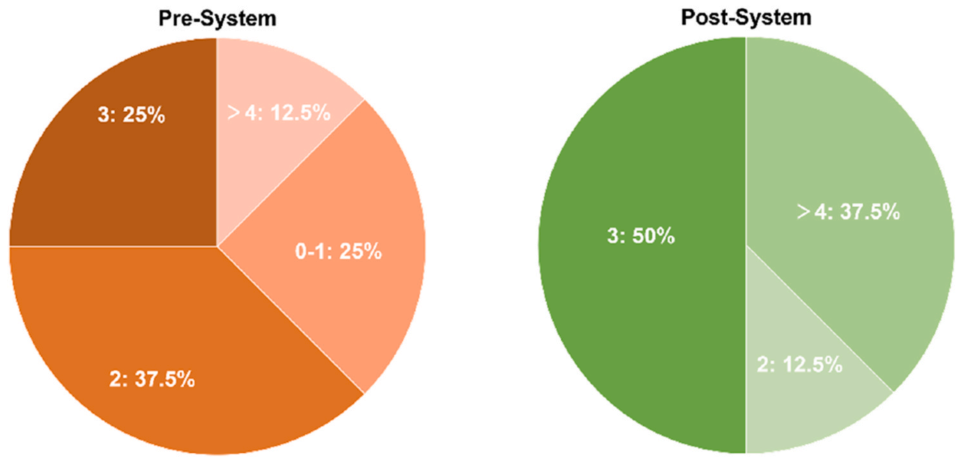

Fig. 4. Improvement in students' ability to identify key friction performance-related variables relevant to their thesis topic before (pre-system) and after (post-system) using the information extraction system. Data were collected from 16 junior students, through pre/post informal interview prompts (Table S1) and system log analysis. 0-1, 2, 3, and >4 in the two panels indicate that students can identify 0-1, 2, 3 and more than 4 influencing variables in the frictional performance of ionic liquids.

technical terms and providing brief tutorials on how to interpret extracted variables for experimental design.

4.3. Interview findings

Informal semi-structured interviews provided deeper insights into the influence of the system on research-based learning. Senior students emphasized that the system helped them organize scattered data into coherent knowledge maps and quantify patterns across studies that would be hard to manually compare. Interviews with junior students (see Tables S1–S2 for question lists) suggested a steeper perceived gain, whereby the visualization tools and standardized outputs reduced their initial confusion with terminology and helped them to recognize core variables more quickly (e.g., surface type, counterface material, and ionic liquid structure). One junior student noted: “Without the system, I would not even know where to start, but with the structured tables I could immediately see the main factors that matter.”

Supervisors supported these impressions. One supervisor remarked: “This tool reduces the cognitive noise for students and the instructional repetition for us.” Another supervisor observed that their consultation sessions shifted from basic explanation to higher-order interpretation, as students arrived with filtered datasets and preliminary comparisons. Students and supervisors identified areas for improvement, such as providing built-in tooltips for technical terms and offering step-by-step examples of how extracted variables inform experimental design.

Alongside these benefits, interviews also surfaced cautionary reflections about how the system should be used. Some students noted that the normalized tables and comparison views can create a sense of completeness, making it tempting to rely on extracted entries without re-checking boundary conditions and measurement context in the original paper. Supervisors similarly emphasized that the system should be treated as a structuring and indexing tool rather than an authoritative evidence source, and recommended verifying any values intended for experimental design or citation through the traceability pointers and the “Confirm” link (Fig. 2). Participants suggested that additional trust-calibration supports (e.g., clearer verification prompts, optional confidence cues for low-certainty fields, and reminders to record missing conditions) would help novices benefit from efficiency gains while avoiding overreliance.

4.4. Observed impact on research-based learning

Beyond user satisfaction, the information extraction system meaningfully impacted students' engagement with research and the development of STEM competencies. The integration of literature-structuring tools not only accelerated information processing but also shifted students' learning from passive consumption to active construction. Rather than merely summarizing the content of documents, students began drawing comparisons, spotting trends, and evaluating methodological designs. These behaviors correspond to higher-order cognitive processes in Bloom's taxonomy (Adams, 2015; Bloom, 1956), such as analysis and evaluation, indicating growth in epistemic cognition.

Students, particularly those researching experimental topics, exhibited marked improvements in their ability to translate literature data into experimental logic. Several junior students demonstrated substantial gains in their ability to recognize variable relationships, such as the effect of counterface material or ionic liquid structure on nano-frictional properties, even without prior background in the field. This finding reflected the effectiveness of the system as a scaffold for inquiry-based learning, enabling novice researchers to engage in authentic reasoning earlier than would be facilitated by traditional instruction alone.

The system also contributed to more confident and independent student attitudes with students able to formulate sharper research questions, justify their methodological choices with empirical references, and compare literature data that reflected analytical rigor rather than superficial summary. Supervisors observed that students using the system posed more thoughtful questions, showed greater initiative, and engaged in deeper discussion during weekly meetings. Such outcomes highlight immediate benefits for thesis completion and longer-term gains in students' readiness to engage in disciplinary research practices, including comparing evidence across sources, articulating methodological rationales, and discussing results with supervisors using shared research artifacts. These findings suggest that the AI-powered information extraction system can not only enhance students’ ability to handle complex scientific literature but also foster critical research competencies that are foundational to higher education and transferable across disciplines.

5. Discussion

The findings of the study suggest that an AI-powered information extraction system can serve as an effective scaffold for supporting undergraduate research-based learning in data-intensive domains. The system helped students transition from passive readers of scientific texts to active constructors of structured knowledge by addressing key challenges faced in undergraduate thesis completion, such as limited

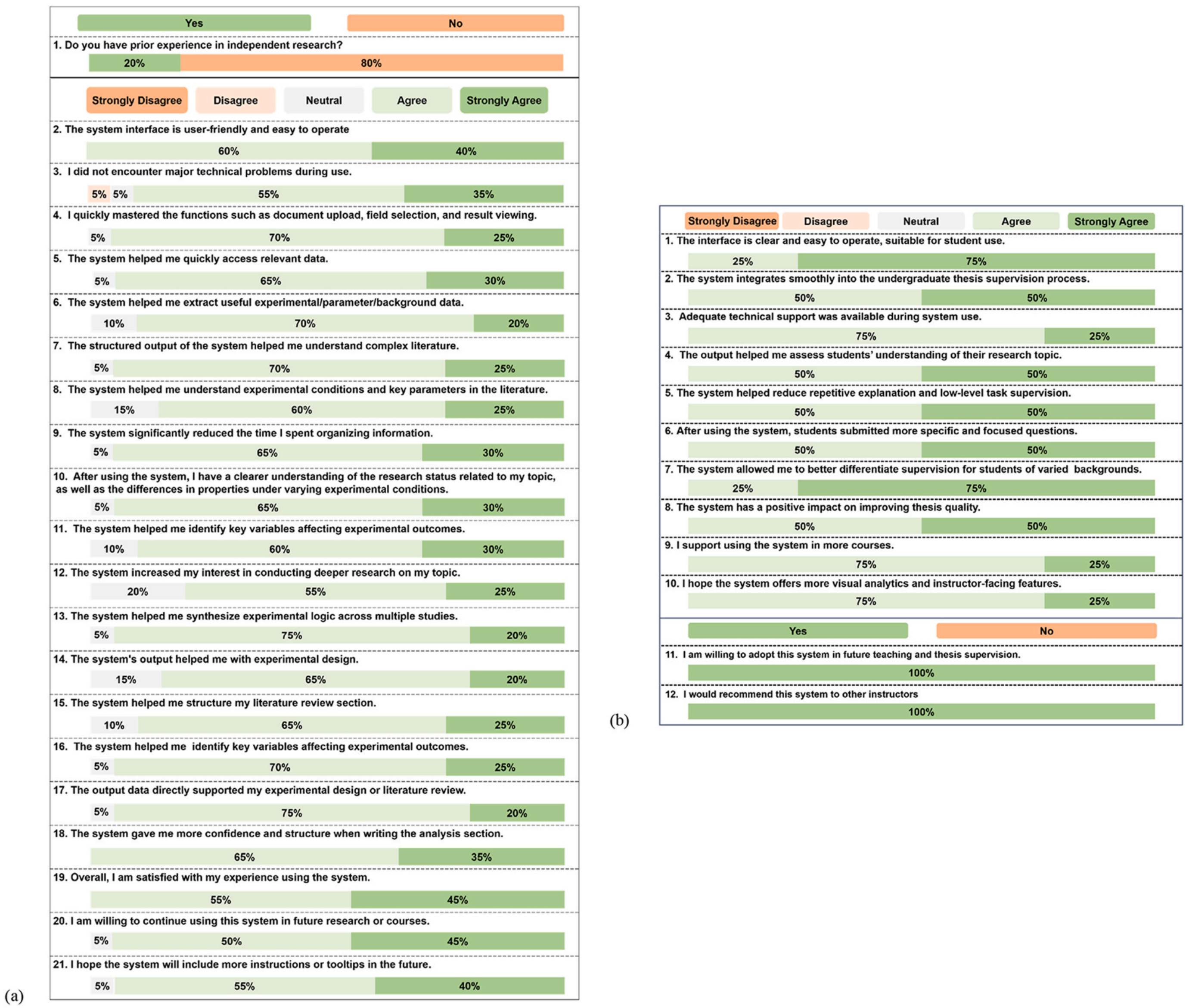

Fig. 5. Analysis of online survey results from (a) the 20 participating students, (b) 4 supervisors who were supervising the 20 participating students.

timeframes, insufficient training, and high cognitive load in literature analysis. This shift resonates with constructivist pedagogy (Hmelo-Silver et al., 2007; Morris, 2025) and the epistemic cognition development, where students increasingly view knowledge not as fixed facts but as evidence-based constructs subject to evaluation and comparison.

From the perspective of educational theory, the findings of the study can be interpreted through three complementary lenses. First, epistemic cognition (Chinn et al., 2011) suggests that thesis-based learning improves when students evaluate how claims are supported by evidence. The traceable source links and structured comparison features of the system support such epistemic work by enabling the inspection of evidence claim relationships. Second, cognitive load theory (Sweller, 1988; van Merriënboer & Sweller, 2005) helps to explain the observed efficiency and reasoning gains among students. By automating low-level search, transcription, and unit standardization, the system reduces students' extraneous cognitive load and frees their cognitive resources for germane processing, i.e., analysis, comparison, and justification. Third, digital scaffolding and inquiry-based learning (Hmelo-Silver et al., 2007) emphasize that tools should guide learners' sense-making rather than replace it. Accordingly, the system is positioned as a scaffold that shifts students' efforts from information gathering toward evidence-based interpretation and decision making.

Unlike generic AI tools that mainly support summarization or text generation, the proposed system is designed to support learning through structured comparison, verification, and traceable evidence use. This distinction underscores the importance of aligning AI design with pedagogical goals, ensuring that technology functions as a cognitive scaffold that promotes research literacy rather than as a shortcut that risks bypassing critical reasoning. A common concern with AI-assisted tools is automation bias, i.e., users may accept generated outputs without critical evaluation (Abubakar et al., 2025), particularly under time pressure, high workload, or when the interface conveys high apparent authority. Recent review cautions that overreliance on AI tools may weaken critical engagement if verification routines are not embedded into instruction (Zhai et al., 2024). In our study, this concern motivated the design stance that traceability and verification should be routine: extracted entries are linked to page/table locations, raw contexts are inspectable, and corrections can be recorded through the editable interface. Together with the manual extraction routine described in Section 3.3, these features position the system as an epistemic scaffold that supports evidence-based interpretation rather than replacing students' engagement with primary literature.

The cautionary reflections in interviews further suggest that traceability features alone may not fully prevent overreliance, because structured outputs can reduce vigilance and shift attention away from boundary conditions unless verification is actively prompted and socially reinforced in supervision. This aligns with prior work on trust in automation, which argues that appropriate reliance depends on calibrating perceived system capability to actual system limitations (Lee & See, 2004). Therefore, beyond making sources accessible, future iterations should strengthen verification scaffolds at key decision points: (i) lightweight uncertainty communication (e.g., optional confidence cues or check-the-source prompts for low-certainty fields), (ii) verification before export/citation (e.g., requiring users to open the evidence location for key records), and (iii) scaffold fading routines that gradually shift responsibility from template-guided extraction toward student-defined field choices and evidence appraisal.

The ability of the system to support cross-document extraction of dynamically updated data and structured visualization contributes to a more transparent, repeatable, and interpretable research process for undergraduate students. In this sense, the system complements rather than competes with bibliographic search platforms, such as Web of Science, by converting retrieved full-text documents into structured, inspectable, and comparable evidence for inquiry and supervision, thereby bridging literature discovery and research-based learning. Pedagogically, effective use of the system requires instructional orchestration rather than tool adoption alone. The following practices are recommended when integrating the proposed system into thesis supervision: 1) verification routines, where students spot-check a subset of extracted entries against original sources and document discrepancies; 2) reflective prompts that require students to justify why selected variables are theoretically relevant and how missing data or boundary conditions may affect interpretation; and 3) scaffold fading, in which early stages provide stronger templates and guidance, while later stages gradually increase students' responsibility for defining extraction fields, evaluating evidence quality, and synthesizing arguments. These strategies align AI support with learning goals and reduce the risk of overreliance on automation.

The feedback from supervisors further highlighted a pedagogical dimension, with students' use of the system reducing instructional redundancy while enhancing differentiated guidance. By allowing advanced students to progress independently and novice students to benefit from structured entry points, the system supports adaptive supervision and personalized learning pathways (du Plooy et al., 2024), aligning with current calls in STEM education for more inclusive and flexible pedagogical designs. The key innovation of this study is the reconceptualization of undergraduate thesis supervision as a data-informed, scaffolded learning process rather than a purely mentor-driven or outcome-focused activity. The system externalizes intermediate research artifacts (e.g., extracted variables, comparison tables, traceable source links, and workflow traces) enabling supervisors to provide earlier, more targeted feedback and supporting differentiated guidance. This process-oriented supervision model is particularly valuable for novice researchers working under constrained timelines, because it shifts the supervision focus from repeatedly explaining basic concepts to supporting evidence-based interpretation and decision making. This reframing is consistent with formative assessment principles (Fitzgerald, 2025), as intermediate research artifacts and traces facilitate timely feedback and iterative improvement. From a learning analytics perspective, prior research cautions that dashboards alone do not guarantee learning gains and typically require complementary pedagogical actions (e.g., facilitation, interpretation routines, and feedback) to become instructionally meaningful (De Laet et al., 2020; Schwendimann et al., 2017). In this study, the learning analytics dashboard offers a lightweight mechanism for data-informed supervision. For example, high iteration counts or repeated incomplete extractions can indicate conceptual uncertainty about theoretically relevant variables. Conversely, stable completion patterns and increasingly targeted searches may signal epistemic progress. Importantly, we used these indicators to prompt supervision dialogue and formative feedback rather than as punitive measures. Such process-level analytics can support formative assessment, differentiated supervision, and early intervention without requiring complex institutional learning analytics infrastructures. Nevertheless, responsible use of the system requires transparency, consent, and avoiding punitive interpretations of trace data. A related pedagogical implication is related to citation sufficiency. The system should be treated as a structuring and indexing tool, not as an authoritative knowledge source. At the record level, one value corresponds to one source, and each extracted entry remains linked to its original publication and evidence location. Students making higher-level synthesis claims supported by multiple studies (e.g., trends across conditions) are encouraged to cite multiple representative primary sources and/or provide a comparison table in which each included record retains its corresponding reference. This practice makes the evidentiary basis explicit and reduces the risk that structured outputs are perceived as citation substitutes rather than as traceable gateways to primary literature.

While this study focused on a specific application scenario in materials science, the system architecture is designed for interdisciplinary transferability, which is crucial for promoting STEM competencies across diverse educational contexts. By adjusting entity templates, field definitions, and domain-specific ontologies, the system can be adapted to support literature-driven learning in other fields where cross-source comparison and data-based reasoning are essential, such as chemistry, biomedical engineering, environmental policy, or even economics and legal research. Cross-domain deployment, e.g., to chemical engineering, biomedical engineering, environmental policy, or economics, remains a next step and will require domain-specific construct mapping, template calibration, and validation with representative corpora. The interdisciplinary adaptability is crucial for promoting STEM competencies, including data literacy, analytical reasoning, and inquiry-based problem solving, across diverse educational contexts.

Finally, the value of the system arguably extends beyond undergraduate thesis completion. Its modularity and data standardization capabilities make it suitable for undergraduate research training projects, innovation and entrepreneurship programs, and course-based design assignments, in addition to graduate-level research training and interdisciplinary laboratory courses. In this sense, the system not only addresses the immediate challenges of undergraduate thesis completion but offers a scalable framework for embedding inquiry-driven, data-informed learning throughout STEM curricula. For early-stage graduate students, the system can serve as a preparatory tool to quickly map state-of-the-art techniques, identify benchmarking metrics, and understand prevailing hypotheses in their target research fields. As an example, the system was utilized by a team leader in the "Challenge Cup" National Undergraduate Curricular Academic Science and Technology Works Competition, known as the Chinese college students of academic science and technology "Olympic" event. Their project focused on the intelligent screening and design of ionic liquid lubricants for extreme conditions. The team was able to rapidly retrieve and organize the state of the art in ionic liquid lubrication using the AI-powered information extraction system, which in turn provided a structured database to support the development of machine learning models for intelligent lubricant design. The project won the Grand Prize (top award) at the provincial level and was later awarded a First Prize at the national level competition. Furthermore, the project was featured in a provincial television news report, underscoring the broader applicability and societal impact of the proposed system. This example illustrates that use of the system can extend beyond thesis preparation to support high-level undergraduate research training, innovation-oriented education, and publicly recognized student achievements. By bridging the gap between literature comprehension, methodological design, and data-informed reasoning, the system contributes to a broader rethinking of how AI can be harnessed as a catalyst for STEM education reform.

6. Limitations and future improvements

While the proposed system yielded promising outcomes in terms of usability, learning impact, and supervision efficiency, several limitations were identified during its deployment, highlighting opportunities for future enhancements. From a technical perspective, extraction accuracy can vary depending on the quality and format of source documents. Scanned PDFs, figures with overlapping labels, or unconventional data representations can occasionally lead to partial or inaccurate field recognition. Although manual correction is supported in the proposed system, integrating an automated flagging or confidence-scoring mechanism could further improve the reliability of extracted results, especially for novice users. In terms of pedagogical application, the system assumes a basic level of data literacy and topic familiarity. While senior students quickly adapted to using the system, several junior students reported initial confusion about how to interpret certain output fields or link data trends to research decisions. This finding suggested the need for complementary instructional materials, such as embedded tooltips, introductory tutorials, or instructor dashboards, to integrate the system into courses with diverse learner backgrounds. Moreover, the system is currently tailored to research tasks in materials science, particularly performance-oriented research areas. Extending the system to disciplines with different knowledge structures, such as the social sciences, law, or design, would require substantial adjustments in field templates, semantic extraction models, and visualization modes. Nevertheless, the modular architecture of the system makes such adaptation feasible in future iterations. A further limitation is that the empirical evaluation was conducted within a single STEM research domain, and full cross-domain porting and validation are warranted in pilot studies that apply the same pipeline with newly defined constructs, field templates, and normalization rules. Furthermore, the current learning analytics dashboard provides primarily descriptive pedagogical analytics (e.g., counts, completion patterns, and time-on-task indicators) rather than predictive learning analytics. Analytics involving longitudinal and sequence-based analyses would better capture students' workflow progression. Finally, a separate performance-based assessment of students' unaided manual data-extraction skills was not performed, warranting an evaluation of workflow transferability to independent extraction and critical appraisal tasks when the system is not available.

To address these limitations, future work will prioritize three directions: 1) expansion of specific field libraries and multilingual support to broaden the applicability of the system; 2) the development of personalized guidance features, such as interactive prompts and learning analytics, to help bridge the gap between raw output and educational meaning; and 3) embedding the system into broader instructional sequences across curricula, supporting thesis completion and other stages of STEM education, including research training courses, innovation projects, and proposal development. By addressing these constraints and leveraging user feedback, the system holds the potential to evolve from a thesis-stage support tool into a scalable infrastructure for data-informed, inquiry-based learning across STEM domains and beyond.

And the pilot study involved a small, single-institution sample with purposive recruitment. Therefore, the findings should be interpreted as exploratory and mechanism-oriented rather than statistically generalizable. Since its initial deployment (Class of 2024), the system has been adopted by an additional cohort of four senior thesis students (2025-2026), and multicohort log data and learning evidence are being collected for a larger-scale evaluation in future work. Future multi-cohort deployments will enable more robust statistical inference (e.g., longitudinal or multi-level analyses).

Finally, most perception measures (survey and interview accounts) were collected post-task, which limits our ability to model changes over time and to disentangle intervention effects from stable individual factors (e.g., prior knowledge, motivation, or research experience). We thus treat the present findings as exploratory and triangulate self-reports with in-task system traces. Future work will adopt a multi-cohort longitudinal design with baseline and repeated measures (e.g., pre/post performance assessments of data extraction and evidence-based justification), enabling analyses that explicitly account for individual differences. In addition, although verification was emphasized in onboarding and supported by traceability links, we did not quantify participants' verification behavior (e.g., frequency of using the "Confirm" link or the proportion of records each student cross-checked). Future work will incorporate these verification actions as explicit log indicators to model trust calibration and to evaluate overreliance-mitigation routines.

7. Conclusion

This study examined the design, implementation, and pedagogical value of an AI-powered information extraction system for supporting undergraduate thesis completion in STEM education. System log analysis, post-task, and semi-structured interviews indicated that the system reduced students' extraneous cognitive load, improved the efficiency and fidelity of literature-driven data handling, and supported the development of research literacy and epistemic cognition (e.g., identifying variables, comparing methods, and justifying design choices). Pedagogically, the system repositions the literature review as evidence organization and comparison rather than mere summarization; while reducing the need for repeated instruction and enabling differentiated guidance, grounded in traceable research artifacts. Methodologically, the mixed-methods design provides a transparent account of process and perception while revealing limitations that inform design iterations.

Specifically, this study articulates an educational innovation in thesis supervision by embedding AI-supported extraction into a structured, process-oriented research-learning workflow. Furthermore, the findings of the study demonstrate that the system lowers the barrier to research engagement by enabling students to identify and compare key experimental variables more effectively. Finally, the study provides evidence from a case study with 20 undergraduate students, reporting measurable gains in efficiency, accuracy, and research reasoning.

Notably, the modularity of the system supports its principled transfer to other STEM domains, provided that construct mapping and template calibration are deliberately undertaken. To avoid overreliance on automation and illusions of completeness, future iterations will integrate context-sensitive guidance (e.g., tooltips and just-in-time prompts), uncertainty/confidence indicators, and reflective checkpoints (e.g., prompts to record missing data and boundary conditions). Embedding the system within research training courses, proposal studios, and innovation programs can extend its impact beyond thesis completion and create a coherent pathway from literature comprehension to data-informed design and evaluation. The system positions AI not as a shortcut but as a scaffold that is explicitly aligned with inquiry and cognition, offering a promising direction for the use of structure-aware AI tools to make expert research practices visible, learnable, and assessable at scale.

CRediT authorship contribution statement

Rong An: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Weian Zhu:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Qian Zhao:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Data curation. **Aatto Laaksonen:** Writing – review & editing, Conceptualization. **Si Lan:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Statements on open data, and ethics

This study was approved by the Academic Committee of Nanjing University of Science and Technology, with ID: 2024103001. Informed consent was obtained from all participants, and their privacy rights were strictly observed. The data can be obtained by sending request e-mails to the corresponding author. Participation was voluntary and unrelated to course grading or thesis evaluation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.caeai.2026.100591.

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